

XUE Boyang

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Research Interests: Trustworthy LLMs/Agent (Uncertainty, Reliability, Knowledge, Hallucinations), Dialogue Systems, Speech Recognition.



Education

The Chinese University of Hong Kong, Hong Kong SAR, China.

Aug. 2021 - Present

Ph.D. Candidate in Department of System Engineering and Engineering Management.

Huazhong University of Science and Technology, Wuhan, China.

Sep. 2016 - Jun. 2020

B.Eng. in School of Artificial Intelligence and Automation, Excellent Class.

Main Publications

Reliable LLM

- **Boyang Xue**, Qi Zhu, Hongru Wang, ..., Qun Liu, Kam-Fai Wong. “ReliableMath: Benchmark of Reliable Mathematical Reasoning of Large Language Models”, Under Review of AAAI 2026 .
- **Boyang Xue**, Fei Mi, Qi Zhu, Hongru Wang, Rui Wang, ..., Xuming Hu, Kam-Fai Wong. “[UAlign: Leveraging Uncertainty Estimations for Factuality Alignment on Large Language Models](#)”, ACL 2025.
- **Boyang Xue**, Hongru Wang, Rui Wang, ... Wenxuan Zhang, Kam-Fai Wong. “[MlingConf: A Comprehensive Study of Multilingual Confidence Estimation on Large Language Models](#)”, ACL 2025 (Findings).

LLM Self-Training

- **Boyang Xue**, Qi Zhu, Hongru Wang, Rui Wang, Sheng Wang, ..., Qun Liu, Kam-Fai Wong. “DAST: Difficulty-Aware Self-Training for Large Language Models”, preprint.

Tool Learning

- Hongru Wang*, **Boyang Xue***, Baohang Zhou, ..., Kam-Fai Wong. “[Self-DC: When to retrieve and When to generate? Self Divide-and-Conquer for Compositional Unknown Questions](#)”, NAACL 2025 (Oral).
- Hongru Wang, Rui Wang, **Boyang Xue**, Heming Xia, Jingtao Cao, ..., Kam-Fai Wong. “[AppBench: Planning of Multiple APIs from Various APPs for Complex User Instruction](#)”, EMNLP 2024.

Parameter Efficient

- Sheng Wang, **Boyang Xue**, Jiacheng Ye, Jiyue Jiang, Liheng Chen, Lingpeng Kong, Chuan Wu. “[PRoLoRA: Partial Rotation Empowers More Parameter-Efficient LoRA](#)”, ACL 2024.

Dialogue System

- Hongru Wang, **Boyang Xue**, Baohang Zhou, Rui Wang, Fei Mi, ..., Kam-Fai Wong. “[UniRetriever: Multi-task Candidates Selection for Various Context-Adaptive Conversational Retrieval](#)”, COLING 2024.
- **Boyang Xue**, Weichao Wang, Hongru Wang, Fei Mi, ..., Kam-Fai Wong. “[Improving Factual Consistency for Knowledge-Grounded Dialogue Systems via Knowledge Enhancement and Alignment](#)”, EMNLP 2023 (Findings).

Speech Recognition

- **Boyang Xue**, Shoukang Hu, Junhao Xu, Mengzhe Geng, Xunying Liu, Helen Meng. “[Bayesian Neural Network Language Models for Speech Recognition](#)”, TASLP (SCI & JCR Q1), 2022.
- **Boyang Xue**, Jianwei Yu, Junhao Xu, Shansong Liu, Shoukang Hu, Zi Ye, Mengzhe Geng, Xunying Liu, Helen Meng, “[Bayesian Transformer Language Models for Speech Recognition](#)”, in ICASSP 2021.

Others

- Rui Wang*, Hongru Wang*, **Boyang Xue***, Jianhui Pang, ..., Heng Ji, Kam-Fai Wong. “Harnessing the Reasoning Economy: A Survey of Efficient Reasoning for Large Language Models”, preprint.
- **Boyang Xue**, Cheng Cheng, Ye Yuan. “[Deep Learning based Patient-Specific FHR Detection System on FECG](#)”, Chinese Patent [CN112826513B].

Research Experiences

- ◆ **Visiting Researcher in Web Intelligence Group, AI Centre, UCL.** May. 2025 - Present
Research Topic: Reliable LLM/Agent. Advisor: [Prof. Aldo Lipani](#) and [Prof. Emine Yilmaz](#)
- ◆ **Ph.D. Student in MoE Lab of High Confidence Software Technologies, CUHK.** Sep. 2022 - Present
Research Topic: Reliable LLM, Dialogue System, etc. Advisor: [Prof. WONG Kam-Fai](#)
- ◆ **Research Assistant and Ph.D. Student in HCCL Lab, CUHK.** Sep. 2020 - Aug. 2022
Research Topic: Bayesian Learning, Speech Recognition. Advisor: [Prof. Xunying LIU](#)
- ◆ **Research Intern in IMDS Lab, HUST.** Sep. 2019 - Jun. 2020
Research Topic: Medical Deep Learning, Patient-specific Diagnosis. Advisor: [Prof. Ye YUAN](#)
- ◆ **Leader in Intelligent Robotics Team, Qi Ming College, HUST.** Nov. 2017 - Aug. 2019
Research Topic: Intelligent Cars, Embedded Systems. Advisor: *Prof. He Ding Xin*

Awards & Honors

- ◆ Excellent Bachelor Thesis, Excellent Student Cadre, Excellent Graduate Student.
- ◆ Academic Scholarship of *Huazhong University of Science and Technology* in 2018-19 academic year.
- ◆ National Grand and First Prize in the 14th and 13th NXP Cup National University Intelligent Car Race.
- ◆ Second Prize in the 19th Asia and Pacific Mathematical Contest in Modeling.

Extracurricular Activities

- ◆ Serving as a reviewer of *ICASSP, ACL, EMNLP, NAACL, COLING, etc.*
- ◆ Serving as a co-founder and organizer of a [NLP Academic Exchange Platform \(NICE\)](#).
- ◆ Serving as a part-time author to write academic blogs on WeChat officials and [Zhihu](#).
- ◆ Serving as a contributor in several open-source projects of [Datawhale](#).
- ◆ Served as the secretary of the Student Union in HUST (2017 - 2019).